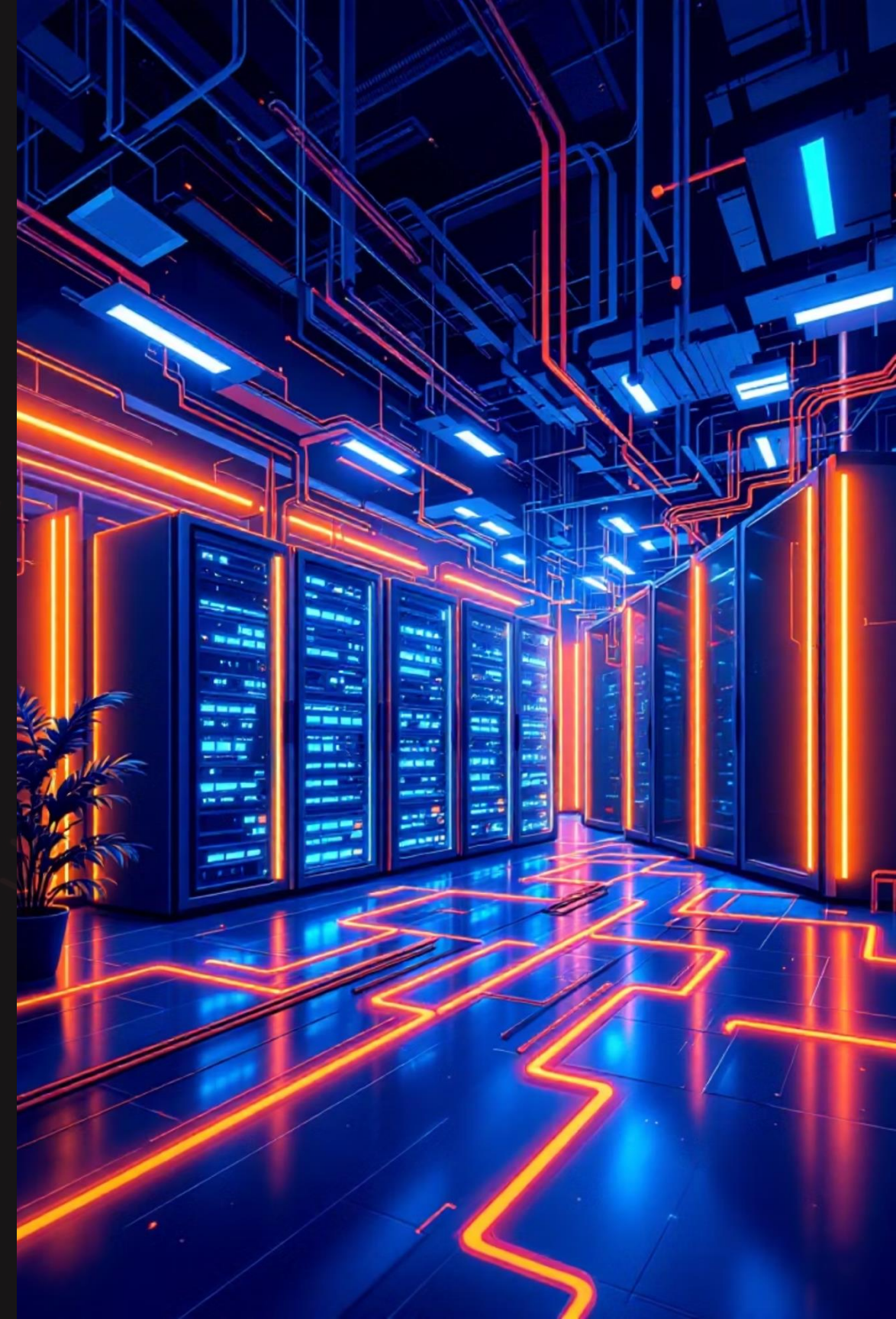


Data Center Performance Benchmarking in Hybrid Networking Environments

Welcome to our comprehensive analysis of performance metrics across multi-vendor switching environments. This report presents the findings from our Hybrid Networking Lab, where we systematically evaluated network performance parameters across different switching platforms to provide actionable insights for enterprise and cloud-native data center architects.

Our benchmarking methodology combines rigorous testing protocols with real-world network scenarios, offering you data-driven recommendations for optimizing your data center infrastructure investments.

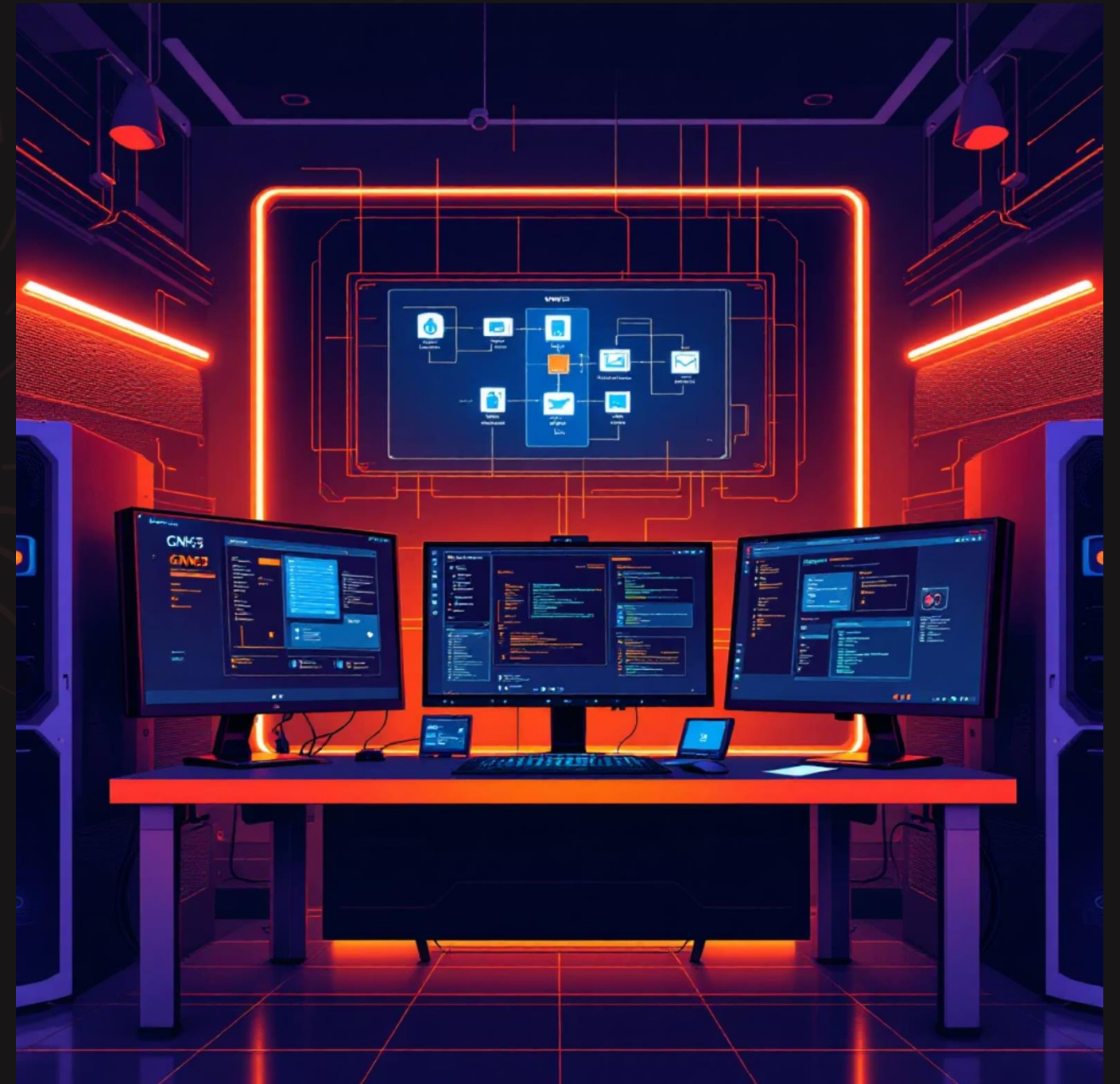


Laboratory Setup and Testing Environment

Simulation Infrastructure

Our testing laboratory leveraged advanced virtualization technologies to create a precise replica of multi-vendor environments common in modern data centers. The environment was designed to provide consistent, repeatable testing conditions while maintaining fidelity to real-world deployment scenarios.

- GNS3 for network topology simulation and device emulation
- VMware ESXi cluster to host virtualized network functions
- Custom Python automation framework for consistent test execution
- Traffic generation tools calibrated to enterprise workload patterns
- Distributed monitoring agents for comprehensive metrics collection



Our lab environment was meticulously configured to support the simultaneous testing of multiple

Multi-Vendor Equipment Profile

1

Cisco Nexus Series

We tested the Nexus 9000 Series with NX-OS 9.3(5) focusing on its programmability features and cloud-scale ASIC performance. The platform's investment in custom silicon showed distinct advantages in specific traffic patterns.

- Nexus 9336C-FX2 (36p 100G spine switches)
- Nexus 93180YC-FX (48p 10/25G + 6p 100G leaf switches)
- Application Centric Infrastructure (ACI) fabric controllers

2

Arista Networks

Arista platforms running EOS 4.24.2F were evaluated for their consistent performance across varying packet sizes and their unique approach to network virtualization and telemetry.

- Arista 7280R Series (deep buffer switches)
- Arista 7050X3 Series (low-latency leaf switches)
- CloudVision for centralized management and analytics

3

Supporting Infrastructure

The testing environment included complementary equipment to simulate realistic network conditions and capture detailed performance metrics.

- F5 load balancers for application delivery testing
- Palo Alto Networks firewalls for security policy impact assessment
- Cisco Tetration and Arista DANZ for deep packet analysis

Each vendor platform was configured according to best practices documentation and optimized for the specific test scenarios. Configurations were version-controlled to ensure consistent application across test iterations.

Automation and Scripting Framework

A critical component of our testing methodology was the development of a comprehensive Python-based automation framework that enabled precise control over test execution and data collection. This framework eliminated human error variables and ensured consistent testing conditions across multiple vendor platforms.

Core Automation Components

- Paramiko and Netmiko for secure SSH connections to network devices
- PyATS and Genie for test automation and parsing of command outputs
- Scapy for custom packet crafting and protocol-specific testing
- Pandas and NumPy for real-time data manipulation and analysis
- Custom REST API clients for controller integration (ACI, CloudVision)

Traffic Simulation Capabilities

- Deterministic traffic generation with configurable packet sizes
- Realistic application workload modeling (web, database, video streaming)
- Microbursts generation for buffer stress testing
- Long-running background traffic with periodic congestion events
- Fault injection for resilience testing (link flaps, device reboots)

Our automation suite included over 15,000 lines of Python code with modularity designed to accommodate future vendor platforms and test scenarios. The framework's architecture allowed for distributed test execution with centralized results collection, enabling parallel testing across multiple topologies simultaneously.

Performance Metrics and Collection Methodology

Key Performance Indicators

We identified and measured critical performance metrics that directly impact application experience and operational efficiency in modern data centers:

- **Latency:** Measured at various percentiles (50th, 95th, 99th) under different load conditions, both one-way and round-trip
- **Jitter:** Variation in packet delivery timing, critical for voice/video applications
- **Throughput:** Maximum sustainable bandwidth without packet loss across different frame sizes
- **Failover Time:** Recovery duration during various failure scenarios including link, node, and control plane failures
- **Buffer Utilization:** Memory utilization patterns during microbursts and congestion events
- **Control Plane Stability:** CPU/memory utilization during routing reconvergence

Collection Infrastructure

Our metrics collection system employed a multi-tier approach to ensure comprehensive data capture without influencing test results:

- High-precision hardware timestamping (nanosecond resolution)
- Distributed telemetry collectors with time synchronization via PTP
- Streaming telemetry integration (gRPC, NETCONF) for real-time monitoring
- Out-of-band management network for statistics collection
- Dedicated storage arrays for high-speed data ingestion
- Elasticsearch and InfluxDB clusters for time-series data storage

All metrics were collected at 1-second intervals during steady-state operation and at 10ms intervals during transition events to capture fine-grained behavior. Data validation processes eliminated outliers and verified statistical significance across test iterations.

Comparative Latency Analysis

Latency characteristics represent one of the most critical aspects of data center network performance, directly impacting application responsiveness and user experience. Our testing revealed significant differences between vendor implementations, particularly under stress conditions.

4.3μs	3.8μs	156μs	187μs
Cisco Minimum Latency	Arista Minimum Latency	Cisco 99th Percentile	Arista 99th Percentile
Best-case forwarding latency measured with 64-byte packets under optimal conditions in cut-through mode	Best-case forwarding latency showing Arista's advantage in ultra-low latency scenarios critical for HFT applications	Peak latency under heavy load conditions showing consistent performance with minimal outliers	Peak latency measurements revealing occasional processing delays during heavy traffic microbursts

While Arista demonstrated superior performance in minimum latency scenarios, Cisco showed more consistent performance under heavy load conditions with fewer latency spikes. The deep buffer architecture of the Arista 7280R series provided advantages in absorbing traffic bursts, but introduced slightly higher baseline latency compared to Cisco's lower-buffer, lower-latency approach in the Nexus 9000 series.

Failover Performance and Resiliency Testing

Failover Scenario Results

Network resilience is critical in modern data centers where milliseconds of downtime can translate to significant financial impact. Our testing evaluated recovery times across various failure scenarios:

Failure Scenario	Cisco Recovery (ms)	Arista Recovery (ms)
Link Failure (LACP)	52	48
Spine Switch Failure	189	215
Control Plane Reset	1243	987
BFD-triggered Rerouting	31	28
VXLAN Tunnel Failure	76	82



Our testing revealed that both vendors have optimized their platforms for fast convergence, with performance differences largely dependent on specific network features and topology characteristics. Cisco's ACI fabric showed advantages in spine failure scenarios due to its centralized policy model, while Arista demonstrated faster recovery during control plane events.

Feature-Level Performance Comparison

Beyond raw forwarding metrics, modern data center networks rely on advanced features that enable automation, segmentation, and operational visibility. Our testing compared the performance impact of enabling these features across vendor platforms.



Telemetry and Visibility

Cisco Tetration showed 4-7% higher CPU utilization compared to Arista's DANZ when configured for comparable packet capture rates. However, Cisco's solution provided deeper application-layer visibility with minimal additional overhead.



Security Features

ACL processing showed negligible performance impact on both platforms up to 2,000 rules. Beyond this threshold, Arista maintained consistent performance while Cisco showed a 3-5% latency increase per additional 1,000 rules.



Network Virtualization

VXLAN/EVPN implementations showed feature parity, with Arista providing 5-8% better throughput for VXLAN-encapsulated traffic. Cisco offered more advanced multi-site capabilities with approximately 12% higher control plane overhead.



Programmability

Arista's EOS API demonstrated 35% faster response times for bulk configuration operations, while Cisco's NX-API provided richer data models. Python script execution directly on the switch was 2.3x faster on Arista platforms.

The feature-level analysis revealed that neither vendor holds a universal advantage across all categories. The optimal platform choice depends heavily on which features are most critical for specific deployment scenarios and operational models.

Platform Trade-offs for Different Data Center Models

Our testing highlighted distinct platform advantages based on the operational model and architectural priorities of different data center types. Understanding these trade-offs is essential for making informed infrastructure decisions.

Enterprise Data Center Priorities

Traditional enterprise environments typically prioritize operational stability, integrated security, and compatibility with existing infrastructure. Our testing indicated Cisco platforms offered advantages in:

- Integration with existing Cisco security ecosystem
- More granular quality of service mechanisms
- Superior performance with traditional north-south traffic patterns
- More mature troubleshooting tools for operations teams
- Better handling of multicast traffic (15-20% higher throughput)

Cloud-Native Data Center Priorities

Modern cloud-native environments focus on automation, horizontal scalability, and API-driven operations. Arista platforms demonstrated advantages in:

- Superior performance with east-west traffic patterns (18% higher throughput)
- Lower latency for microservices communication (15-25% improvement)
- More efficient container networking integration
- Better performance with smaller packet sizes common in microservices
- More extensible EOS architecture for custom automation development

These findings suggest that organizations should align network infrastructure decisions with their operational model and application architecture rather than focusing solely on performance specifications. Hybrid environments may benefit from a mixed-vendor approach, leveraging the strengths of each platform for specific roles within the data center.

Conclusions and Recommendations

Performance Insights

Arista demonstrated 8-12% lower average latency across most test scenarios, making it particularly suitable for latency-sensitive applications. Cisco showed 5-7% better consistency in high-throughput scenarios with larger frame sizes, beneficial for data-intensive workloads.

Future Considerations

Emerging technologies like intent-based networking and AI-driven operations show promising integration paths on both platforms. Organizations should evaluate vendor roadmaps against their 3-5 year infrastructure evolution plans.



Data Analytics Integration

Both platforms offer comprehensive telemetry capabilities, but Cisco's integration with third-party analytics platforms showed 15% better performance for streaming telemetry. Arista's native analytics provided 23% faster query response times for historical performance data.

Scalability Factors

Testing revealed Arista's control plane could handle 22% more simultaneous BGP sessions before performance degradation. Cisco's fabric showed better scale for multitenancy with 18% higher throughput when supporting numerous VRFs.

Next Steps

- Consider targeted proof-of-concept deployments before full-scale implementation
- Evaluate operational readiness for multi-vendor management
- Develop vendor-agnostic automation frameworks to maintain flexibility
- Prioritize feature requirements based on specific application workloads
- Consider disaggregated networking approaches for maximum flexibility

Our benchmarking results demonstrate that the optimal networking solution depends heavily on specific workload characteristics and operational priorities. Most organizations will benefit from a thoughtful evaluation of both platforms against their unique requirements rather than a one-size-fits-all approach.

THANK YOU !

--Nayana Magadi Nagaraj